Table 55. Maximum ADC R_{AIN} (continued)

Resolution	Sampling cycle at 35 MHz	Sampling time at 35 MHz (ns)	Max. R_{AIN}[1] (Ω)
6 bits	1.5	43	390
	3.5	100	2200
	7.5	214	5600
	12.5	357	10000
	19.5	557	15000
	39.5	1129	33000
	79.5	2271	50000
	160.5	4586	50000

1. Specified by design. Not tested in production.

Table 56. ADC accuracy

Symbol	Parameter[1][2]	Conditions	Min	Typ	Max	Unit
ET	Total unadjusted error	$V_{DDA} = V_{REF+} = 3$ V $f_{ADC} = 35$ MHz, $f_s \leq 2.5$ Msps, $T_A = 25°C$	-	±3	±4	LSB
		$2\text{ V} < V_{DDA} = V_{REF+} < 3.6$ V $f_{ADC} = 35$ MHz, $f_s \leq 2.5$ Msps, T_A = entire range	-	±3	±6.5	
EO	Offset error	$V_{DDA} = V_{REF+} = 3$ V $f_{ADC} = 35$ MHz, $f_s \leq 2.5$ Msps, $T_A = 25°C$	-	±1.5	±2	LSB
		$2\text{ V} < V_{DDA} = V_{REF+} < 3.6$ V $f_{ADC} = 35$ MHz, $f_s \leq 2.5$ Msps, T_A = entire range	-	±1.5	±4.5	
EG	Gain error	$V_{DDA} = V_{REF+} = 3$ V $f_{ADC} = 35$ MHz, $f_s \leq 2.5$ Msps, $T_A = 25$ °C	-	±3	±3.5	LSB
		$2\text{ V} < V_{DDA} = V_{REF+} < 3.6$ V $f_{ADC} = 35$ MHz, $f_s \leq 2.5$ Msps, T_A = entire range	-	±3	±5	
ED	Differential linearity error	$V_{DDA} = V_{REF+} = 3$ V $f_{ADC} = 35$ MHz, $f_s \leq 2.5$ Msps, $T_A = 25$ °C	-	±1.2	±1.5	LSB
		$2\text{ V} < V_{DDA} = V_{REF+} < 3.6$ V $f_{ADC} = 35$ MHz, $f_s \leq 2.5$ Msps, T_A = entire range	-	±1.2	±1.5	
EL	Integral linearity error	$V_{DDA} = V_{REF+} = 3$ V $f_{ADC} = 35$ MHz, $f_s \leq 2.5$ Msps, $T_A = 25$ °C	-	±2.5	±3	LSB
		$2\text{ V} < V_{DDA} = V_{REF+} < 3.6$ V $f_{ADC} = 35$ MHz, $f_s \leq 2.5$ Msps, T_A = entire range	-	±2.5	±3	
ENOB	Effective number of bits	$V_{DDA} = V_{REF+} = 3$ V $f_{ADC} = 35$ MHz, $f_s \leq 2.5$ Msps, $T_A = 25$ °C	10.1	10.2	-	bit
		$2\text{ V} < V_{DDA} = V_{REF+} < 3.6$ V $f_{ADC} = 35$ MHz, $f_s \leq 2.5$ Msps, T_A = entire range	9.6	10.2	-	